

About the Data

Spatial Data Sets

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The PurpleAir data used in the California case study were processed prior to upload and analysis in *AQ-Shiny*. The preprocessing described here is specific to the preparation of the case-study data and is not performed within *AQ-Shiny*. The application assumes that users have conducted appropriate quality-control and preprocessing procedures for their data before upload.

The PurpleAir data were downloaded in 2020 for outdoor sensors located throughout California. The downloaded data covered the 2019 calendar year; however, the case study presented in this manuscript uses observations from January 2019. At the beginning of January 2019, 1,015 PurpleAir sensors were available across California. Following the preprocessing and selection procedures described below, the spatial case-study data set used in *AQ-Shiny* contained 829 sensor locations.

PurpleAir sensors contain two Plantower PMS5003 particle sensors, referred to as channels A and B, that operate during alternating 10-second intervals and provide repeated measurements of particulate matter concentrations. Prior to May 30, 2019, PurpleAir measurements were reported as 80-second averages; subsequently, measurements were reported as 2-minute averages (Barkjohn et al., 2020). The Plantower sensors report estimated mass concentrations of PM_1 , $PM_{2.5}$, and PM_{10} . PurpleAir data also include measurements of temperature and relative humidity. The $PM_{2.5}$ concentrations used in the *AQ-Shiny* case study are expressed in $\mu g/m^3$.

Quality-control procedures were applied to the high-frequency PurpleAir measurements before temporal aggregation. These procedures were informed by the PurpleAir quality-assurance and data-cleaning criteria described by Barkjohn et al. (2020). PurpleAir measurements occasionally contain extreme temperature values resulting from electrical noise or communication errors between the temperature sensor and the PurpleAir microcontroller. Accordingly, observations containing implausible temperature values were removed prior to aggregation. Observations with relative humidity values outside the physically valid range of 0–100% were also excluded.

The two Plantower sensors within each PurpleAir unit were used to evaluate the consistency of the $PM_{2.5}$ measurements. Following the approach described by Barkjohn et al. (2020), agreement between channels A and B was evaluated using both the absolute concentration difference and the percent difference between the two channels. The percent difference was calculated as

$$\text{Percent Difference} = \frac{2(A - B)}{A + B} \times 100,$$

where A and B denote the $PM_{2.5}$ concentrations reported by channels A and B, respectively. Measurements were considered inconsistent when the two channels differed by more than $5 \mu\text{g}/\text{m}^3$ and by more than 61%, and observations meeting both criteria were excluded. This combined criterion avoids flagging small absolute differences at low $PM_{2.5}$ concentrations solely because they correspond to large percentage differences (Barkjohn et al., 2020).

After quality control and evaluation of agreement between channels A and B, valid measurements from the two channels were combined and aggregated to obtain hourly $PM_{2.5}$ concentrations for each sensor. The hourly relative humidity measurements were prepared at the same temporal resolution.

The hourly PurpleAir measurements were subsequently corrected using the U.S.-wide correction developed by Barkjohn et al. (2020),

$$PM_{2.5, \text{corrected}} = 0.524, PA_{\text{cf1}} - 0.0862, RH + 5.75,$$

where PA_{cf1} denotes the hourly averaged $PM_{2.5}$ concentration from the `cf_1` measurements of channels A and B, and RH denotes the corresponding hourly relative humidity expressed as a percentage. The corrected $PM_{2.5}$ concentrations are expressed in $\mu\text{g}/\text{m}^3$ and were used as the response variable in the AQ-Shiny case study.

Barkjohn KK, Gantt B, Clements AL (2020). Development and application of a United States-wide correction for $PM_{2.5}$ data collected with the PurpleAir sensor. *Atmospheric Measurement Techniques*.

File Explanation

- `spatial_only.csv`: Data set used for modeling
- `pred_locs_spatial.csv`: Data set used for prediction

Spatio-Temporal Data Sets

The regulatory-monitoring data used for the second *AQ-Shiny* case study were obtained from the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS). The source data consist of hourly $PM_{2.5}$ measurements from monitoring sites located throughout California

during the 2019 calendar year. The data set covers January 1 through December 31, 2019 and contains measurements from 77 unique monitoring sites.

The $\text{PM}_{2.5}$ measurements correspond to AQS parameter and are reported in $\mu\text{g}/\text{m}^3$. The monitoring records in the case-study source data are classified as Federal Equivalent Method (FEM) measurements.

The AQS data include monitoring-site identifiers, geographic coordinates, local and GMT dates and times, $\text{PM}_{2.5}$ concentrations, measurement units, method information, and associated site information. For the revised *AQ-Shiny* analysis, $\text{PM}_{2.5}$ is modeled on the concentration scale in $\mu\text{g}/\text{m}^3$.

For the *AQ-Shiny* case study, a subset of the processed AQS data from January 1, 2019 through January 7, 2019 on 40 unique monitoring sites is used to demonstrate spatio-temporal visualization, Gaussian process model fitting, prediction, and model comparison. The data are prepared in the input format required by *AQ-Shiny* before upload.

File Explanation

- `st_example.csv`: Data set used for modeling
- `pred_stl.csv`: Data set used for prediction